



Advanced Machine Learning

2023/2024

Sample Exam 2024 Time: 2h00

Name:

Number:

This is a **closed books** exam. Any attempt at cheating implies immediate disapproval in the course and the activation of the procedures defined in the disciplinary regulation document of the UC.

Question #	Points	Your score
1	20	
2	20	
3	20	
4	20	
5	20	
	100	

Grader:

Pergunta 1 - Define 20 points

Define the following concepts in a clear and **rigorous** way. You may use, if applicable, an example to **help you** to explain the concept.

a) Stacking.

Your answer:

b) Denoising Auto-Encoder.

Your answer:

c) Discriminative Model.

Your answer:

d) Large Language Model.

Your answer:

Pergunta 2 - Solve 20 points

You have an ensemble of five classifiers each of them having a certain weight. They were trained to solve a classification problem where there are three possible outcomes. In figure 1 you see the results of testing the ensemble with an unseen instance. We would like to decide what should be the result of the ensemble using the following aggregation methods:

- majority voting
- weight average.

Justify your answer.

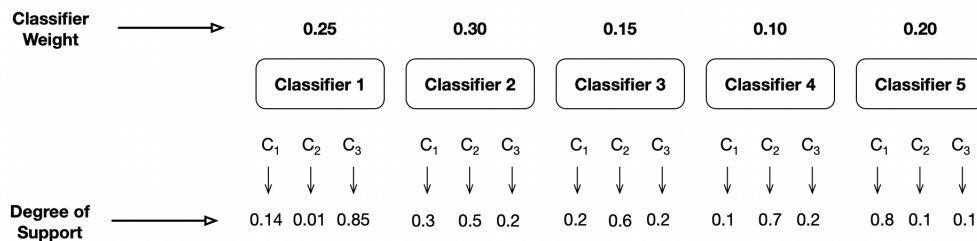


Figure 1: An ensemble at work!

Your answer:

Your answer:

Pergunta 3 - Solve 20 points

Apply the convolutional filter to the feature map and generate the output of the operation according to the following configurations: padding = 0, stride = 2.

Feature Map				Filter/Kernel	
0	0	1	0	1	1
0	1	1	0	1	1
0	0	1	0		
0	0	1	0		

Figure 2: Feature Map and filter.

Your answer:

Your answer:

Pergunta 4 - [Explain](#) 25 points

In Machine Learning, the data is essential to the train and test any learning algorithm. Explain what is data modality and describe the most relevant types of data modality that exist.

Your answer:

Your answer:

Pergunta 5 - Open Problem 25 points

Ruombla hired you for their autonomous robot vacuum cleaner division. The latest model must operate independently based on the following inputs: infrared proximity sensors to sense obstacles, current position mapped into a 2D space, dirt detector and the battery level sensor. The robot is continuously cleaning when operating, can move in any direction by receiving input commands (let us simplify the commands as: move towards point (x,y) in the 2D space, turn 90 degrees), can check its position in the 2D space and can check if there is dirt in front of it. The objective is to develop a solution that learns to control the robot to clean a room (a 2D space) efficiently. Based on what you learned in class, what approach would you take to handle the problem? **Explain** your solution in terms of: **machine learning pipeline, data preparation, learning paradigm, models/architectures and relevant training hyperparameters** (if applicable).

Your answer:

Your answer: